# Microsoft Entra ID, Linux Commands and Identity Management

## 1. Azure Active Directory vs Microsoft Entra ID

Azure Active Directory (Azure AD) was Microsoft's cloud-based identity and access management service. It was mainly used to manage users, applications, devices, and access to cloud resources. Azure AD helped organizations control who could access their applications and resources and provided features such as authentication, single sign-on, and multi-factor authentication.

Microsoft renamed Azure Active Directory to **Microsoft Entra ID** in 2023. The name was changed as part of the Microsoft Entra product family, which focuses on identity and access management. Microsoft Entra ID is not a completely new service. It is essentially the new name for Azure AD, with the same core identity management capabilities continuing under the new branding.

Microsoft Entra ID allows administrators to create and manage users and groups, assign permissions, manage application access, and implement security policies. It can also be integrated with Microsoft 365, Azure resources, and many third-party applications.

Some important features of Microsoft Entra ID include:

- User and group management
- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Application and device management
- Identity protection and security monitoring
- Access policies and conditional access

The main difference between Azure AD and Microsoft Entra ID is therefore mostly **the name and broader product branding**, rather than a completely different technology. Existing Azure AD services were transitioned to Microsoft Entra ID. For example, Azure AD users are now referred to as Microsoft Entra ID users, and Azure AD administrators are generally referred to as Microsoft Entra administrators.

Understanding this change is important because older documentation, tutorials, and training materials may still use the term Azure Active Directory or Azure AD, while newer Microsoft documentation uses Microsoft Entra ID.

## 2. Basics of Linux Commands

Linux is an operating system widely used for servers, cloud computing, networking, and development. Unlike graphical operating systems where many tasks are performed using menus and windows, Linux users often interact with the system through a **command-line interface (CLI)**.

Linux commands allow users to navigate directories, create and delete files, manage permissions, monitor processes, and perform many other administrative tasks.

**pwd** - Displays the current working directory.

**ls** - Lists files and directories in the current location.

**cd** - Changes from one directory to another.

**mkdir** - Creates a new directory.

**touch** - Creates a new empty file.

**cp** - Copies files or directories from one location to another.

**mv** - Moves or renames files and directories.

**rm** - Removes files or directories. Care should be taken because deleted files may not be easily recoverable.

**cat** - Displays the contents of a text file.

**clear** - Clears the terminal screen.

**echo** - Displays text or can be used to write text into files.

**grep** - Searches for specific text or patterns within files or command output.

**chmod** - Changes the permissions of a file or directory.

**whoami** - Displays the username of the currently logged-in user.

**sudo** - Allows an authorized user to execute commands with administrative privileges.

**man** - Displays the manual or documentation for a Linux command.

Learning these commands provides a basic understanding of Linux administration. They are especially useful when working with Linux servers, cloud environments, and cybersecurity-related tasks.

## 3. Types of Identity

Identity refers to the information used to identify a person, device, application, or other entity within a system. Identity management is important because organizations need to know **who or what is requesting access** before allowing access to resources.

### User Identity

A user identity represents an individual person. It normally contains information such as a username, password, email address, and assigned permissions. For example, an employee may have a Microsoft Entra ID account that allows them to access company applications.

### Device Identity

A device identity represents a computer, laptop, mobile phone, virtual machine, or another device. Organizations can use device identities to control which devices are allowed to access their resources.

### Application Identity

An application identity represents an application or software service that needs to access resources. Applications may require identities so that they can authenticate themselves when communicating with other services.

### Service Identity

A service identity is used by a service or automated process to authenticate and access required resources. This is particularly useful for background services and automated tasks that do not involve a human user.

### Group Identity

A group identity represents a collection of users or other identities. Instead of assigning permissions individually to every user, administrators can assign permissions to a group and then add the required users to that group.

Identity management helps organizations improve security by ensuring that users and services receive only the access they actually require.

## 4. Difference Between User, Group and Role

In identity and access management, **users and groups** are commonly used to organize identities and assign access. A role is different because it normally represents a set of permissions or responsibilities.

### User

A user is an individual identity representing a person. Each user generally has their own account and authentication credentials. For example, an organization may have separate accounts for employees such as a manager, developer, or administrator. Permissions can be assigned directly to individual users when necessary.

### Group

A group is a collection of users or identities. Groups make administration easier because permissions can be assigned to the group instead of assigning the same permissions to every user individually. For example, a company could create a group called Developers and add all developers to it. If the Developers group is given access to a particular application, all members of the group can receive that access according to the organization's policies.

Groups are especially useful in large organizations because managing hundreds of individual accounts separately would be inefficient.

### Role

A role defines a set of permissions or administrative responsibilities. Roles are used to determine what actions or permissions an identity is allowed to perform. For example, an administrator role may allow a user to manage other users, while a more limited role may only allow the user to view certain information.

Roles are important for **Role-Based Access Control (RBAC)**. RBAC allows organizations to assign permissions based on job responsibilities rather than giving every user complete access.